

eRCA: A Knowledge-driven Explainable Root Cause Analysis Approach for Software Troubleshooting

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Abstract—Intelligent software operation and maintenance (O&M) is a key process for stable system operation in software engineering, which consistently ensures business requirements and users' expectations even post-delivery. Meanwhile, potential issues resolved during O&M can be integrated into the software development procedure, creating positive feedback that effectively reduces prospective faults as software evolves. However, existing intelligent software troubleshooting methods suffer from low accuracy, low efficiency and poor interpretability, resulting from insufficient analysis of the root causes of software faults. In this paper, we propose a knowledge-driven explainable root cause analysis (eRCA) approach for intelligent software troubleshooting with effective knowledge management and reuse. We design a novel fault representation structure with a portrait model to categorically depict multimodal troubleshooting data. Then, a computational operation and maintenance knowledge graph (COM-KG) is constructed to conduct graph-based root cause analysis, which is further employed for knowledge-driven software troubleshooting. Our evaluation shows the high quality of COM-KG and the eRCA-based method. Accurate, instant and explainable solutions for software troubleshooting are achieved, resulting in lower maintenance resources, significant development inspiration and proactive implementation for end-users.

Index Terms—AI for SE, intelligent software troubleshooting, root cause analysis, computational operation and maintenance knowledge graph

I. INTRODUCTION

Software operation and maintenance (O&M) plays a critical role in the intelligent software development cycle, ensuring stable software system operation in production environments to meet end-user requirements [1]–[3]. Meanwhile, troubleshooting knowledge can be introduced to software development as prior knowledge to avoid potential failures [4], [5]. One major challenge in software troubleshooting is to efficiently identify fault root causes and precisely recommend feasible fault resolution approaches [6]–[8].

In modern industry, research on fault diagnosis techniques conducted fault detection and isolation with mathematical

model-based methods [9]–[11]. Knowledge-based fault resolution approaches provide a highly effective way to manage the complexity of software systems and generate explainable solutions [12]–[14].

Despite the progress achieved by the researchers in the abovementioned work, knowledge-based software troubleshooting faces new issues. First, massive historical O&M data for software systems are usually characterized by multimodal, multi-granularity and multi-dimensional in practical industrial environments [15]–[17]. Existing knowledge representation frameworks fail to provide a systematical schema to depict comprehensive fault information. Second, current research mostly incorporate explicit fault symptoms to match historical cases that occur in similar phenomenon. However, the cases that share same root cause can obtain more insights to efficiently provide essential solutions.

This paper proposes a knowledge-driven explainable root cause analysis (eRCA) approach for software troubleshooting, which enables accurate, efficient and interpretable software troubleshooting [18]–[21]. To realize effective knowledge management and reuse of multimodal root cause information, we propose a novel schema to categorically depict various software faults. To the best of our knowledge, virtually no research provides explainable root cause analysis (eRCA) to generate prescriptive maintenance measures for software troubleshooting. In this paper, we make the following contributions:

- We design a novel multi-granularity fault portrait model to realize accurate fault representation.
- We incorporate knowledge graph as external knowledge for intelligent software fault solution reasoning and root cause analysis.
- A uniform explainable software troubleshooting framework named eRCA is proposed to enhance the software O&M efficiency and software development quality.

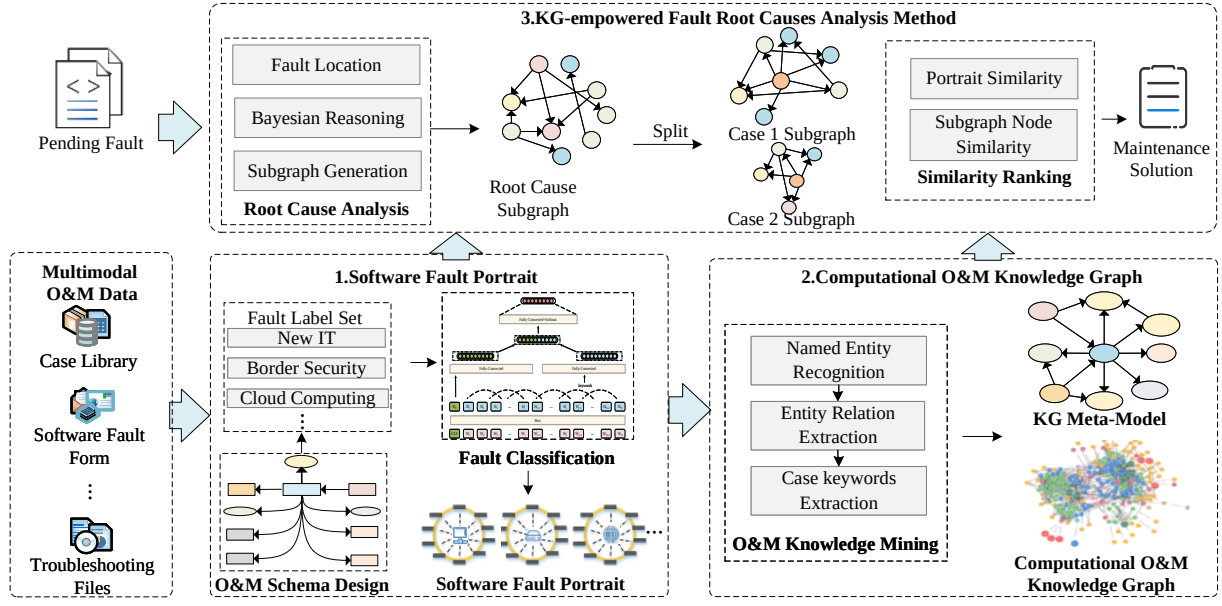


Fig. 1. The overall framework of the knowledge-driven explainable root cause analysis approach for intelligent software troubleshooting. The proposed framework includes three modules: (1) Software Fault Portrait, (2) Computational O&M Knowledge Graph and (3) KG-empowered Fault Root Cause Analysis for Software Troubleshooting.

II. METHODOLOGY

The overall framework of the proposed approach is depicted in Fig. 1. Under this framework, multi-modal O&M data can be categorically modeled as a software fault portrait and managed with a knowledge graph. Consequently, accurate, highly correlated and interpretable O&M case recommendations based on knowledge-driven fault root cause analysis can be achieved. The proposed framework includes three modules: (1) Software Fault Portrait, (2) Computational O&M Knowledge Graph, and (3) KG-empowered Fault Root Cause Analysis for Software Troubleshooting.

A. Software Fault Portrait

Fault cases comprise multi-dimensional information such as devices, modules, operations, and symptoms. Each element demonstrates semantic relationships such as causality and inclusion, which are crucial for accurate maintenance solutions. Existing fault case analysis methods are conducted based on keyword extraction, hindering the adaptation to complex O&M knowledge in practical scenarios. Inspired by the concept of user portrait, we introduce the construction of software fault portraits to enable fault case modeling under diverse conditions.

The knowledge contained in fault cases consists of description information and processing solutions, which collectively provide a comprehensive understanding of the faults. Consequently, the O&M portrait schema proposed in this paper includes five components, namely basic information, device information, content information, impact information, and attribute information. Specifically, basic information describes fault case details such as date and keywords. Device information depicts essential information about the faulty software, including software type, model, and manufacturer.

Content information is subdivided into description information and processing information, containing a significant amount of fault entities and related information. Impact information is separated into the severity level and the probability of occurrence to collectively represent the resultant impact. Attribute information encompasses details including fault number, fault type, and keywords, which serve as a basic supplement to the fault content.

B. Computational O&M Knowledge Graph

Representing the O&M information with a knowledge graph at a high semantic level can provide an explicit representation and enhance the precision of case suggestions through knowledge association. Moreover, the use of KG for intelligent O&M facilitates intuitive and interpretable recommendation outcomes for users.

Entity-relation Extraction. To address the challenges of error accumulation, information redundancy and interaction missing in existing pipeline extraction models, we propose a joint entity-relation extraction model based on a pointer network. Our joint extraction model improves the accuracy and efficiency of the entity-relation extraction, while also providing a more comprehensive understanding of the correlations between entities. BiLSTM model is utilized to encode head entities and capture the dependency features of words within the sentence at varying distances.

Keywords Extraction. To capture the global information of the case and extract important fault keywords, this paper enhances the YAKE model [22] and proposes a heuristic keyword extraction algorithm. We first perform word separation and lexical annotation on the sentences and constrain the word separation based on manually constructed fault external dictionaries. Then, the corresponding feature scores are calculated

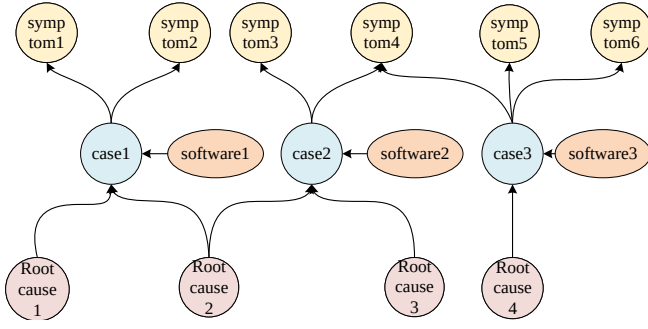


Fig. 2. The DAG structure of fault O&M data. Since a fault case may contain multiple root cause entities and symptom entities, there exists a one-to-many relationship $\langle \text{fault ID}, \text{root causes} \rangle$ and $\langle \text{fault ID}, \text{symptom} \rangle$.

by analyzing the word frequency, lexicality, location and other features. N-gram expansion is incorporated according to the sliding window, and finally a set of keyword lists is generated with duplicates removed through relevance and score sorting. The formula for calculating the candidate keyword score is as follows.

$$S(\text{keyword}) = \frac{\prod_{t \in \text{keyword}} S(t)}{KF(\text{keyword}) \times \left(1 + \sum_{t \in \text{keyword}} S(t)\right)} \quad (1)$$

C. KG-empowered Fault Root Cause Analysis Method

Traditional fault case recommendation methods are often based on matching similar cases according to explicit symptoms. However, the correlation between cases with the same root causes is often ignored, making it difficult to provide accurate recommendations for fault resolution. Thus, we propose a KG-empowered fault root cause analysis method and employ the approach for O&M case recommendations. The proposed method incorporates KG-based reasoning and portraits matching together to generate recommended O&M cases.

Bayesian Network Construction. Bayesian network (BN) is represented as $G = (G_D, \gamma)$, where $G_D = (N, E)$ is the directed acyclic graph structure (DAG). N stands for the set of nodes, and E is the set of edges. $\gamma = \{P(N_i | pa(N_i))\}$ is the set of conditional probability, where $pa(N_i)$ is the parent node set of node N_i . The DAG structure of fault O&M data is shown in Fig. 2.

By leveraging the semantic knowledge within the COM-KG, the node information of the fault title and the problem description is recorded as symptom nodes in the DAG. The faulty devices in COM-KG are transformed into system nodes in the DAG, while root cause entities are converted to root cause nodes. Then, directed edges are generated between $\langle \text{root cause node}, \text{fault node} \rangle$, $\langle \text{system node}, \text{fault node} \rangle$ and $\langle \text{fault node}, \text{symptom node} \rangle$ to establish directional relationships.

Root Cause Subgraph Generation. Based on the fault node location and root cause analysis, the root cause subgraph can be generated according to the candidate entities and underlying relations on COM-KG. For different relation types, entity

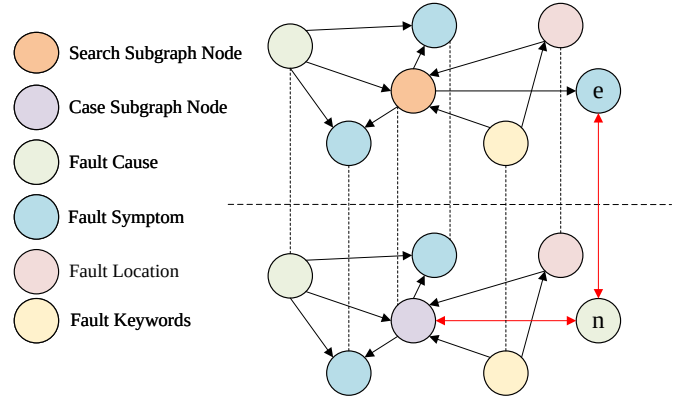


Fig. 3. The process of multi-hop neighbor association of root cause subgraph.

types and entity locations, different queries are constructed to complete the query of the candidate edges. Additionally, new entities may appear in the retrieval process and are added to the subgraph.

For further computation for root cause subgraph similarity, the constructed subgraph can be split into multiple cases based on case ID. The fault case ID is retrieved over COM-KG based on the head and tail entities and relations associated. We emphasize that the head and tail entities as well as relationships must be present in the case simultaneously. Finally, multiple related case subgraphs are generated based on respective case IDs.

Similarity based on Portrait Matching. We first conduct portrait-based similarity calculation. Portrait matching incorporates the software fault portraits constructed from historical O&M data and performs the retrieval of the input faults. First, we utilize TF-IDF and the categorical weight of entities to represent the fault portraits. By calculating the cosine similarity between the portrait embedding of recall cases and the portrait representation of input cases, a recommendation list is generated.

For each entity e in fault portraits, the TF-IDF value is calculated by the following formula:

$$\Pi(e) = \frac{\sum_{e \in \text{case}} \text{num}(e)}{\sum_{u \in E_{\text{case}}} \text{num}(u)} \times \log \left(\frac{\sum_{c \in \text{case}} \text{num}(\text{case})}{\sum_{c \in \text{case} \cap e \in c} \text{num}(\text{case}) + 1} \right) \quad (2)$$

Similarity based on Root Cause Subgraph Nodes. Considering the incorporation of the neighborhood information of root cause subgraphs, multi-hop relationships among nodes in the subgraphs are introduced to include the information of entity nodes in the search subgraph that are not currently accounted for in the calculation. The process of multi-hop neighbor association is shown in Fig. 3.

As shown in Fig. 3, retrieve paths within two hops between node e and node c on COM-KG Path_e is represented as $\{p_1, p_2, \dots, p_n\}$. For m entities, the path set of subgraph difference nodes is denoted as $\{\text{Path}_1, \text{Path}_2, \dots, \text{Path}_m\}$. Then, the similarity of the case root cause subgraph is computed as:

$$S_{sim} = \frac{1}{|\text{Path}|} \times \sum_{j=1}^m t_j F(|\text{Path}_j|) \quad (3)$$

$$C_{sim} = \lambda P_{sim} + (1 - \lambda) S_{sim} \quad (4)$$

In summary, the O&M case relevance is calculated by integrating the similarity of portrait matching and the root cause subgraph in formula (4), where λ is the weight of portrait similarity and is chosen as 0.85 after performing corresponding experiments.

III. EVALUATION

In this section, we report the evaluation of our eRCA-based method to answer the following three research questions:

RQ1: What is the intrinsic quality of RCA information in COM-KG?

RQ2: How effective is the COM-KG based eRCA?

RQ3: How is the quality of software troubleshooting based on the eRCA method?

A. Dataset Construction

To evaluate the proposed method, we collected nearly 10,000 fault cases from an open-access online BBS forum provided by Sangfor¹. Each case contains modules such as title, description information, alarm information, processing solution, root cause and so on. Each module consists of multi-modal information including pure texts, troubleshooting forms, and corresponding images. We filtered the special characters and kept the pure texts of different modules. Finally, a total of 7,176 valid cases were collected. These cases were constructed as the original dataset.

To construct the dataset for entity-relation extraction, we chose BERT-wwm as our pre-trained language model to extract triplets from the original dataset. OpenKG² and real triplets extracted from product documents were selected as the external knowledge to introduce prior features. In total, we collected 17,254 samples containing 40,937 triplets. The dataset was split into train/test sets by 7 : 3.

B. Intrinsic Quality of RCA Information in COM-KG (RQ1)

The quality of RCA Information in COM-KG essentially depends on the accuracy of information extraction algorithms, which aims to build multi-granularity software fault RCA information. According to the process of the extraction, we conducted experiments in entity-relation extraction.

We evaluated our entity-relation extraction method with the rule-based model and the tagging model. Our proposed model was trained using *Adam* optimizer and then tested on the test set. To measure the quality of extraction results, precision(P), recall(R) and F_1 -score were calculated, with all samples in the test set treated equally. The main results of different models are shown in Table I.

It can be observed that the proposed model achieves the best results compared with the baselines. Besides, our model does not need to define the types of entities, which require less data and can significantly reduce the upfront cost. For ablation tests, it can be observed that the employment of prior features,

TABLE I
THE RESULTS FOR EXTRACTION ALGORITHMS ON THE ORIGINAL DATASET.

Models	P (%)	R (%)	F_1 (%)
Rule-based Model	47.23	42.39	44.68
Tagging Model	64.76	61.93	63.31
BERT	65.76	62.93	64.31
BERT+prior features	67.64	65.58	66.59
BERT+BiLSTM	67.06	64.85	65.96
BERT+self-attention	67.41	65.12	66.25
BERT+prior features+self-attention	69.92	68.33	69.16
BERT+BiLSTM+self-attention	69.17	68.25	68.71
Joint Extraction Model	71.82	70.76	71.29

TABLE II
THE RESULTS OF ROOT CAUSE ANALYSIS ON DIFFERENT TEST SETS

Test Set	Top-k	N	Accuracy		Error Rate		
			N_{top-k}	Acc	T	ϵ_T^a	P_ϵ
Test Set 1	1	100	54	0.54	574	46	0.08
	3	100	67	0.67	574	99	0.17
	5	100	81	0.81	574	95	0.17
	8	100	83	0.83	574	136	0.24
Test Set 2	1	50	17	0.34	263	33	0.13
	3	50	25	0.50	263	75	0.29
	5	50	33	0.66	263	85	0.32
	8	50	36	0.72	263	112	0.43
Average	1	150	71	0.47	837	79	0.09
	3	150	92	0.61	837	174	0.21
	5	150	114	0.76	837	180	0.22
	8	150	119	0.79	837	248	0.30

^a T is the number of actual nodes.

ϵ_T is the number of wrong predicted nodes.

BiLSTM encoding and self-attention respectively improves F_1 by 3.28%, 2.65% and 2.94%. These improvements show that these modules have a significant effect on the accuracy of joint extraction of entity-relations.

C. Quality of COM-KG based Knowledge-driven eRCA(RQ2)

For RQ2, we randomly selected 100 cases from the original dataset as test set 1. Then we randomly selected 50 cases and rewrote them by synonyms replacement as test set 2. Among them, the cases in test set 1 exist in the KG constructed above. While test set 2, as a new dataset, has different entities and relations, while the root causes remain the same. Such settings can effectively evaluate the adaptation of our eRCA method on unseen faults.

The effectiveness of root cause analysis is verified by judging whether the actual causes appear in the inference root causes nodes. Another metric, accuracy (Acc), is used in this section. Accuracy is calculated by the percentage of cases whose actual root causes appear in the inference nodes. For the wrong cases, we calculate the error rate(P_ϵ) by the percentage of wrong inference nodes.

We chose Top-k nodes respectively as 1, 3, 5 and 8 for experiments. The results on different test sets are shown in

¹<https://bbs.sangfor.com.cn/>

²<http://www.openkg.cn/>

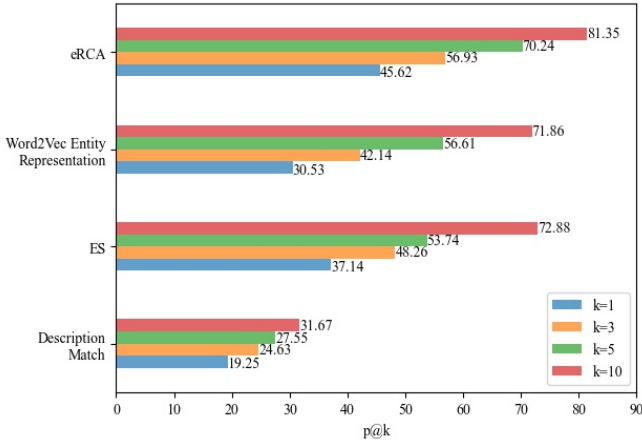


Fig. 4. The results of different algorithms for solution recommendation. Our model outperforms the Word2Vec Entity Representation method, which shows the importance of knowledge.

Table II. The accuracy on test set 1 is higher than that on test set 2 since the cases in test set 1 are already stored in the KG. These cases will match multiple similar nodes including the original ones, so the accuracy can reach 83%, proving that our proposed Bayesian reasoning-based root cause analysis algorithm has a strong inference capability. Our model achieves an accuracy of 72% on test set 2, which laterally proves that our algorithms can still maintain a good performance encountering unexpected entities and relations. On average, our proposed model achieves an accuracy of 79%, which shows its effectiveness.

D. Quality of eRCA-based Software Troubleshooting (RQ3)

We constructed the solution recommendation test dataset based on manual selection. For each of the 6 fault types, we selected 20 base cases and constructed 10 similar cases each. Similar cases are (1) collected from the historical fault case libraries, or (2) rewritten by synonyms replacement as described in Section IV C. For each case in the case list, we manually calculated its similarity to the base case. Finally, the ranking results of different markers are combined to obtain a ranking list of similar cases.

Our eRCA-based method uses the description of each case as the input, and chooses the top k cases as the recommendation results. In order to evaluate the overall performance, we use the following baselines.

- **Description Match:** calculate the similarity based on the description information of fault cases.
- **Elasticsearch (ES):** use invert index for matching. This method can associate cases based on duplicate fields in the text.
- **Word2Vec Entity Representation:** use the average of word vectors getting from Word2Vec to represent entities instead of KG.

The main results are shown in Fig. 4. We can observe that the description match achieves the poorest results since there is too much interference information in the description. On the other hand, using the sentence vector of BERT directly

TABLE III
THE RESULTS OF DIFFERENT RANKING METHODS

Models	P (%)	R (%)	F_1 (%)
Portrait-based matching	50.44	59.68	54.67
Subgraph-based matching	54.75	62.53	58.38
Combined matching	67.38	73.46	70.29

achieve lower accuracy. The accuracy is obviously improved using ES, which proves that there is less effective information in descriptions again. Our proposed model considers fault portrait and root cause subgraphs of the fault cases, which outperforms other baseline methods. Meanwhile, *Word2Vec* entity representation achieves lower accuracy than that based on knowledge representation, which proves the importance of the knowledge-empowered model for semantic information extraction. In order to verify the effectiveness of the ranking method, we use the P , R and F_1 , comparing with the manual ranked case lists. Table III shows the comparison of different methods. Our proposed recall method improves the F_1 to 70.29%, significantly better than the above methods. This shows that our model successfully combines both the text and association information.

IV. CONCLUSION AND FUTURE WORK

This paper presents an automatic approach for performing explainable root cause analysis based software troubleshooting. The approach is built on a novel fault portrait representation of implicit troubleshooting knowledge extracted from multimodal historical O&M data. To conduct effective knowledge management and discovery, we construct a knowledge graph named computational O&M graph (COM-KG) that allows graph-based root cause analysis for unexpected faults. Based on root cause reasoning, we propose a solution recommendation method with both portrait-based and subgraph-based matching. We collected 7,176 software fault samples for evaluation and built our COM-KG with 40,937 triplets extracted. Our evaluation indicates that eRCA can provide high-quality fault knowledge and effective software troubleshooting based on knowledge-driven explainable root cause analysis. In the future, we will apply the eRCA-based software troubleshooting method for internal software troubleshooting technical support on a well-known multinational information and communication technology (ICT) company. Besides, the enhancement will be made with more advanced graph-based methods such as policy-guided path reasoning for explainable recommendations.

V. DATA AVAILABILITY

Both our dataset and source code can be accessed on our anonymous repository³ for future research endeavors.

³<https://anonymous.4open.science/r/eRCA-C728>

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